

Deterministic Optimization

Large-Scale Optimization

Cutting Stock Problem

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Column Generation

Large-Scale Optimization

Learning Objectives for this lesson

- Discover an algorithm to solve an LP with a large number of variables, called column generation

Cutting Stock Problem

- The Cutting Stock Problem is formulated as
 - $\min \sum_{i=1}^N x_i$
 - s.t. $\sum_{i=1}^N A_i x_i = b$
 - $x \geq 0$
 - where the columns A_i of A are the patterns for how to cut a large roll
 - b is the amount of demand of each size of smaller rolls
 - N is an extremely large number
 - There is no way we have the patience or time to write down all the columns of A

Column Generation Algorithm

We want to solve a large-scale LP with many variables

1. Pick a subset I of columns $\{A_j\}$ and variables x_j for $j \in I \subseteq \{1, 2, \dots, N\}$
2. Solve the so-called restricted master problem

$$\begin{aligned} (MP) \quad & \min \sum_{j=1}^N c_j x_j \\ & \text{s.t.} \quad \sum_{j=1}^N A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j = 1, \dots, N. \end{aligned}$$

Original LP with many variables

$$\begin{aligned} (RMP) \quad & \min \sum_{j \in I} c_j x_j \\ & \text{s.t.} \quad \sum_{j \in I} A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j \in I. \end{aligned}$$

Restricted LP with much fewer variables

Variables $\Leftrightarrow$ Columns

Key Property 1 of RMP

- A feasible solution of RMP can be made into a feasible solution of MP
 - If $\bar{x}_j$ for $j \in I$ is a solution of (RMP)
 - Then, set all other variable $x_j = 0$ for $j \notin I$
 - The resulting solution is feasible for (MP)
 - Why? Because (RMP) has all the constraints in (MP).

$$\begin{aligned} (RMP) \quad & \min \sum_{j \in I} c_j x_j \\ & \text{s.t.} \quad \sum_{j \in I} A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j \in I. \end{aligned}$$

$$\begin{aligned} (MP) \quad & \min \sum_{j=1}^N c_j x_j \\ & \text{s.t.} \quad \sum_{j=1}^N A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j = 1, \dots, N. \end{aligned}$$

Key Property 2 of RMP

- A basic feasible solution of RMP can be made into a basic feasible solution of MP
 - If $\bar{x}_j$ for $j \in I$ is a BFS of (RMP)
 - Then, set all other variable $x_j = 0$ for $j \notin I$
 - The resulting solution is BFS for (MP)
 - Why? Because (RMP) has all the constraints in (MP), then the basis matrix of (RMP) is also a basis matrix of (MP)
 - So the basic variables of (RMP) are also basic variables of (MP)

$$\begin{aligned} (RMP) \quad & \min \sum_{j \in I} c_j x_j \\ & \text{s.t.} \quad \sum_{j \in I} A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j \in I. \end{aligned}$$

$$\begin{aligned} (MP) \quad & \min \sum_{j=1}^N c_j x_j \\ & \text{s.t.} \quad \sum_{j=1}^N A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j = 1, \dots, N. \end{aligned}$$

Key Property 3 of RMP

- For an optimal BFS of (RMP), we can compute reduced cost of all nonbasic variables, if any reduced cost is negative, then we know the optimal solution of (RMP) is not optimal for (MP)
- We can add the new variable with negative reduced cost to (RMP), solve the new (RMP), and repeat the process
- The procedure of finding a variable with negative reduced cost is called the **Pricing Problem**

$$\begin{aligned} (RMP) \quad & \min \sum_{j \in I} c_j x_j \\ & \text{s.t.} \quad \sum_{j \in I} A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j \in I. \end{aligned}$$

$$\begin{aligned} (MP) \quad & \min \sum_{j=1}^N c_j x_j \\ & \text{s.t.} \quad \sum_{j=1}^N A_j x_j = b \\ & \quad x_j \geq 0, \quad \forall j = 1, \dots, N. \end{aligned}$$

Column Generation Algorithm

(Cont.)

3. Pricing Problem:

- Goal: Compute all the reduced costs of $\bar{x}$. If all reduced costs are nonnegative, then $\bar{x}$ is optimal for (MP); otherwise, we find a new column to add to (RMP).
- Key idea: To achieve this, compute the **minimum** reduced cost. If the minimum reduced cost is nonnegative, then all reduced costs ≥ 0
 - Compute $w = \min_{j=1,\dots,N} \{c_j - \mathbf{c}_B^T \mathbf{B}^{-1} \mathbf{A}_j\}$
 - If $w \geq 0$, then all reduced costs are nonnegative. Terminate column generation.
 - If $w < 0$, pick the column j with the minimum reduced cost, add $\mathbf{A}_j$ to (RMP).

Correctness and Convergence

- Whenever we design a new algorithm, we need to ask the question:
 - Is the algorithm correct?
 - Yes for Column Generation, because of the key properties of (RMP)
- Does the algorithm converge?
 - Yes, because the algorithm always adds new columns and never disregards any. The worst case is all the columns of (MP) are added to (RMP), then the algorithm will terminate. Otherwise, the algorithm terminates before adding all the columns (this is usually what happens).

Summary

- We learned:
 - Column Generation Algorithm